

Session 8: Introduction to PL/SQLTrainer Coverage:PL/SQL block structure (DECLARE, BEGIN, EXCEPTION, END).Variables and cursors.Control structures (IF, LOOP).Demo: Simple stored procedure to calculate user's total spend.Pending

Tasks

- 1) Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

Query:

```
Run SQL query/queries on database my database: ⓘ
3 CREATE PROCEDURE show_user()
4 BEGIN
5     DECLARE user_name VARCHAR(50);
6     DECLARE user_age INT;
7
8     SET user_name = 'Abhi';
9     SET user_age = 20;
10
11     SELECT user_name AS 'User Name',
12            user_age AS 'Age';
13 END //
14
15 DELIMITER ;
16
17 CALL show user();|
```

Output:

Extra options	
User Name	Age
Abhi	20

- 2) Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.

Query:

```
Run SQL query/queries on database my database: ⓘ
1 DELIMITER //
2
3 CREATE PROCEDURE check_delivery()
4 BEGIN
5     DECLARE order_amount DECIMAL(10,2);
6
7     SET order_amount = 600;
8
9     IF order_amount > 500 THEN
10         SELECT 'Eligible for free delivery' AS Message;
11     ELSE
12         SELECT 'Delivery charges apply' AS Message;
13     END IF;
14 END //
15
16 DELIMITER ;
```

Output:

```
Extra options
Message
Eligible for free delivery
```

- 3) Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE.

Hint: Use a counter variable and EXIT WHEN condition inside the loop.

Query:

```
Run SQL query/queries on database my database: ?

3 CREATE PROCEDURE print_numbers()
4 BEGIN
5     DECLARE counter INT DEFAULT 1;
6
7     simple_loop: LOOP
8         SELECT counter;
9
10        SET counter = counter + 1;
11
12        IF counter > 5 THEN
13            LEAVE simple_loop;
14        END IF;
15    END LOOP simple_loop;
16 END //
17
18 DEF TMTTER .
```

Output:

Routines ?

☐ Check all

	Name	Type	Returns	
☐	check_delivery	PROCEDURE	Edit Execute Export Drop	
☐	print_numbers	PROCEDURE	Edit Execute Export Drop	
☐	show_user	PROCEDURE	Edit Execute Export Drop	

- 4) Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE.

Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.

QUERY:

Run SQL query/queries on database my database: 

```
1 CREATE TABLE PRODUCTS (  
2     PRODUCT_ID INT PRIMARY KEY AUTO_INCREMENT,  
3     PRODUCT_NAME VARCHAR(100)  
4 );  
5  
6 INSERT INTO PRODUCTS (PRODUCT_NAME) VALUES  
7 ('Laptop'),  
8 ('Mobile Phone'),  
9 ('Keyboard'),  
10 ('Mouse'),  
11 ('Headphones');
```

Run SQL query/queries on database my database: 

```
2  
3 CREATE PROCEDURE show_products()  
4 BEGIN  
5     DECLARE done INT DEFAULT 0;  
6     DECLARE p_name VARCHAR(100);  
7  
8     DECLARE product_cursor CURSOR FOR  
9         SELECT PRODUCT_NAME FROM PRODUCTS;  
10  
11     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;  
12  
13     OPEN product_cursor;  
14  
15     product_loop: LOOP  
16         FETCH product_cursor INTO p_name;  
17
```

```
18  
19     END IF;  
20  
21     SELECT p_name AS Product_Name;  
22  
23     END LOOP;  
24  
25     CLOSE product_cursor;  
26 END //  
27  
28 DELIMITER ;
```

OUTPUT:

Extra options

				PRODUCT_ID	PRODUCT_NAME
☐	Edit	Copy	Delete	1	Laptop
☐	Edit	Copy	Delete	2	Mobile Phone
☐	Edit	Copy	Delete	3	Keyboard
☐	Edit	Copy	Delete	4	Mouse
☐	Edit	Copy	Delete	5	Headphones

Routines

☐ Check all

Export

Drop

	Name	Type	Returns			
☐	check_delivery	PROCEDURE	Edit	Execute	Export	Drop
☐	print_numbers	PROCEDURE	Edit	Execute	Export	Drop
☐	show_products	PROCEDURE	Edit	Execute	Export	Drop
☐	show_user	PROCEDURE	Edit	Execute	Export	Drop

5) Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

QUERY:

```
1 1  
2  
3 CREATE PROCEDURE divide_numbers()  
4 BEGIN  
5     DECLARE num1 DECIMAL(10,2) DEFAULT 10;  
6     DECLARE num2 DECIMAL(10,2) DEFAULT 0;  
7     DECLARE result DECIMAL(10,2);  
8  
9     DECLARE CONTINUE HANDLER FOR SQLSTATE '22012'  
10    BEGIN  
11        SELECT 'Cannot divide by zero' AS Message;  
12    END;  
13  
14    SET result = num1 / num2;  
15  
16    SELECT result AS Result;
```

OUTPUT:

